

	Material Safety Data Sheet	Document Num.	
		Issued Date	2023-10-10
		Revised on	

1. CHEMICAL PRODUCT AND COMPANY INFORMATION

A. Product name

SUPOL HR530U

B. Recommended use of the chemical and restrictions on use

- ☐ Product application: Automotive Exteriors
- ☐ Restrictions on use: Sufficient drying

C. Information on manufacturer/supplier/distributor

- ☐ Manufacturer: HYUNDAI EP co., ltd 1221-32 Daehoman-ro, Seokmun-myeon, Dangjin-si, Chungcheongnam-do, KOREA
- ☐ Supplier: HYUNDAI EP co., ltd
- ☐ Address: 1221-32 Daehoman-ro, Seokmun-myeon, Dangjin-si, Chungcheongnam-do, KOREA
- ☐ Information: +82-(0)31-785-2808
- ☐ Emergency contact phone number: HYUNDAI EP co., ltd / Technical Support Team / Kim Ki Su
(Telephone +82-(0)31-785-2808, Mobile phone +82-(0)10-7676-1294)

2. HAZARD IDENTIFICATION

A. Hazard classification

Not classifiable.

B. Label elements including precautionary statements

- ☐ Symbols: Data not available
- ☐ Signal word(s): Data not available
- ☐ Hazard statement(s): Data not available
- ☐ Precautionary statement(s)
 - ☐ Prevention: Data not available
 - ☐ Response: Data not available
 - ☐ Storage: Data not available
 - ☐ Disposal: Data not available

C. Other hazards which do not result in classification(NFPA)

NFPA Chemical name	Health	Flammability	Reactivity
1. Polypropylene	1	1	0
2. Ethylene-Propylene copolymer	1	1	0

3. 1-Butene, polymer with ethene	1	1	0
4. Talc, Non-asbestis form	1	0	0
5. Second Antioxidant (2,4-DI-TERT-BUTYLPHENOL PHOSPHITE)	1	1	0
6. Hindered Amine Compound (Fatty acids, 2,2,6,6-Tetramethyl-4-Piperidiny Esters)	1	1	0

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical name	Common name, synonym	CAS number	Percentages(%)
Polypropylene	Polypropene	9003-07-0	8~12
Ethylene-Propylene copolymer	Ethylene-Propylene polymer	9010-79-1	50~60
1-Butene, polymer with ethene	Polyolefin elastomer, Ethylene alpha-olefin copolymer	25087-34-7	10~15
Talc, Non-asbestis form	Talc	14807-96-6	15~20
Second Antioxidant (2,4-DI-TERT-BUTYLPHENOL PHOSPHITE)	IRGAFOS 168	31570-04-4	0.1~1
Hindered Amine Compound (Fatty acids, 2,2,6,6-Tetra methyl-4-Piperidiny Esters)	Tetrahydroxybenzophenone	167078-06-0	0.1~1

4. FIRST AID MEASURES

A. Eye contact

- ☐ Flush eyes thoroughly with water for 20 minutes at least.
- ☐ If effects occur, consult a physician.

B. Skin contact

- ☐ Flush eyes thoroughly with water for 20 minutes at least.
- ☐ Remove and isolate contaminated clothing and shoes
- ☐ Clean up clothing and shoes perfectly before reuse
- ☐ Seek medical attention immediately.

C. Inhalation

- ☐ Seek medical attention immediately.
- ☐ Move person to fresh air
- ☐ If not breathing, give artificial respiration
- ☐ If breathing is difficult, give oxygen

D. Ingestion

- ☐ Do not give anything by mouth to an unconscious person
- ☐ Seek medical attention immediately.

E. Indication of immediate medical attention and special treatment needed, if necessary

- ☐ Treatment should be directed at the control of symptoms and the clinical condition of the patient

5. FIRE-FIGHTING MEASURES

A. Suitable extinguishing media

- ☐ Suitable extinguishing media: Dry sand, Dry chemicals, Alcohol-resistant foam, Water spray, Form extinguisher, CO2
- ☐ Unsuitable extinguishing media: high pressure water injection
- ☐ Giant fire: Water spray/fog, Form extinguisher

B. Specific hazards arising from the chemical

- ☐ Hazardous combustion products
 - ☐ Harmful gases could be occurred including CO & CO2
 - ☐ Inhalation of the substance may be hazardous
 - ☐ Some vapors may cause choking and dizziness
- ☐ Fire and explosion hazards
 - ☐ Products could be ignited by heat, sparks and flame
 - ☐ Containers can explode when heated
 - ☐ Some products can burn but does not ignite easily

C. Special protective equipment and precautions for firefighters

- ☐ Move containers from fire area if you are not dangerous
- ☐ Some products may be transported at high temperature
- ☐ The spilled material may cause contamination
- ☐ Upon contact with skin and eyes can cause burn
- ☐ Fire-extinguishing water has to be trapped in a ditch
- ☐ In case of fire on the tank, cool the container off by large amounts of water after putting out a fire
- ☐ In case of fire on the tank, step back from the container if it sounds high-pitched tone from pressure relief device or discolors
- ☐ In case of fire on the tank, step back from container in flames

6. ACCIDENTAL RELEASE MEASURES

A. Personal precautions, protective equipment and emergency procedures

- ☐ Use appropriate safety equipment (skin and eyes protection)

B. Environmental precautions

- ☐ Prevent from entering into waterways, sewers, basements, confined space

C. Methods and materials for containment and cleaning up

- ☐ Small quantity leakage: Wash out a contaminated area as large amount of water
 - ☐ Absorb the leaked material using sand, non-combustible materials and put it on the container
 - ☐ Close a lid loosely after put the leaked materials on a dry container using a clean shovel
 - ☐ and move it from contaminated area to safe area
- ☐ Large quantity leakage: Dig a ditch far away from contaminated area
- ☐ Powder leakage: For prevention the spread of powder, cover a contaminated area with plastic sheets and keep dry

7. HANDLING AND STORAGE

A. Precautions for safe handling

- ☐ Handle a product at normal room temperature, atmospheric pressure and keep away from ignition source

B. Storage for safe

- ☐ Keep separated from incompatible substances stored at normal room temperature, atmospheric pressure
- ☐ Do not store near fire

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

A. Exposure limits in the air of the workplace, biological limit values

1) Polypropylene

- ☐ Domestic regulations: ND
- ☐ ACGIH regulations: ND
- ☐ Biological exposure limits: ND

2) Ethylene–Propylene copolymer

- ☐ Domestic regulations: ND
- ☐ ACGIH regulations: ND
- ☐ Biological exposure limits: ND

3) 1–Butene, polymer with ethene

- ☐ Domestic regulations: ND
- ☐ ACGIH regulations: ND
- ☐ Biological exposure limits: ND

4) Talc, Non–asbestos form

- ☐ Domestic regulations
 - o TWA – 6mg/m³ Soapstone (Total dust)
 - o TWA – 3mg/m³ Soapstone (Respirable dust)
 - o TWA – 2mg/m³ Talc (Asbestos excluded)
- ☐ ACGIH regulations
 - o (Respirable, asbestos excluded), TWA 2 mg/m³
- ☐ Biological exposure limits: Not applicable

5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE)

- ☐ Domestic regulations: ND
- ☐ ACGIH regulations: ND
- ☐ Biological exposure limits: ND

6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–PiperidinyI Esters)

- ☐ Domestic regulations: ND
- ☐ ACGIH regulations: Not Applicable
- ☐ Biological exposure limits: Not Applicable

B. Appropriate engineering controls

- ☐ If fumes or dust may be generated in a work area, provide local ventilation

C. Individual protection measures

- ☐ Respiratory protection
 - o If processing fumes are not adequately controlled, use appropriate respirator for protection from organic vapors

- o Respiratory protector is arranged from minimum to maximum, and consider warning properties before use
- o Available particulate respirators
 - Dust/mist particulate respirators
 - Air-filtration particulate respirators (High-efficiency particulate filter medium)
 - Powered air purifying respirators
 - Self-contained breathing respiratory equipped high-efficiency particulate filter medium
- Eye protection
 - o Wear safety glasses or goggles while using or handling product
 - o Equip the emergency shower and washing facilities near workplace
- Hand protection
 - o Put on appropriate chemical-resistance gloves
- Body protection
 - o Wear appropriate chemical-resistance clothing

9. PHYSICAL AND CHEMICAL PROPERTIES

A. Appearance

- | | |
|------------------|----------------|
| ○ Physical state | Solid (Pellet) |
| ○ Colour | Black |

B. Odour

ND

C. Odour threshold

ND

D. pH

ND

E. Melting point/freezing point

165~170°C

F. Initial boiling point and boiling range

ND

G. Flash point

260 °C / Method – closed system

H. Evaporation rate

ND

I. Flammability(solid, gas)

ND

J. Upper/lower flammability or explosive limits

ND

K. Vapour pressure

ND

L. Solubility(ies)

ND

M. Vapour density

ND

N. Specific gravity

0.96~1.02

O. Partition coefficient: n-octanol/water

ND

P. Auto-ignition temperature

ND

Q. Decomposition temperature

ND

R. Viscosity

ND

S. Molecular weight

ND

10. STABILITY AND REACTIVITY

A. Chemical stability and possibility of hazardous reactivity

- Stable at room temperature and atmospheric pressure
- Irritant and toxic gases may generate in case of fire

B. Conditions to avoid

- ☐ Exposure to elevated temperature, Flame, Ignition source

C. Incompatible materials

- ☐ Avoid contact with oxidizing agents and such as peroxides
- ☐ Combustible materials

D. Hazardous decomposition products

- ☐ Phosphorus oxides , Carbon oxides, Hydrocarbon gases, Hydrocarbones, Aldehyde

11. TOXICOLOGICAL INFORMATION**A. Information on the likely routes of exposures**

- ☐ Inhalation exposure: ND
- ☐ Ingestion exposure: ND
- ☐ Skin exposure: Burn may occur in case of short-term exposure
- ☐ Eye exposure: ND

B. Delayed and immediate effects and also chronic effects from short and long term exposure**1) Polypropylene**

- ☐ Acute toxicity
 - ☐ Oral: LD50 > 8000 mg/kg Rat
 - ☐ Skin: ND
 - ☐ Inhalation: ND
- ☐ Skin corrosion/irritation: ND
- ☐ Serious eye damage/irritation: ND
- ☐ Respiratory sensitization: ND
- ☐ Skin sensitization: ND
- ☐ Carcinogenicity
 - ☐ Occupation safety and health act: ND
 - ☐ Notification of ministry of labor: ND
 - ☐ IARC: Group 3
 - ☐ OSHA: ND
 - ☐ ACGIH: ND
 - ☐ NTP: ND
 - ☐ EU CLP: ND
- ☐ Germ cell mutagenicity: ND
- ☐ Reproductive toxicity: ND
- ☐ Specific target organ systemic toxicity–single exposure: ND
- ☐ Specific target organ systemic toxicity–repeated exposure: ND
- ☐ Aspiration hazard: ND

2) Ethylene–Propylene copolymer

- ☐ Acute toxicity
 - ☐ Oral: ND
 - ☐ Skin: ND

- o Inhalation: ND
- ☐ Skin corrosion/irritation: ND
- ☐ Serious eye damage/irritation: ND
- ☐ Respiratory sensitization: ND
- ☐ Skin sensitization: ND
- ☐ Carcinogenicity
 - o Occupation safety and health act: ND
 - o Notification of ministry of labor: ND
 - o IARC: ND
 - o OSHA: ND
 - o ACGIH: ND
 - o NTP: ND
 - o EU CLP: ND
- ☐ Germ cell mutagenicity: ND
- ☐ Reproductive toxicity: ND
- ☐ Specific target organ systemic toxicity—single exposure: ND
- ☐ Specific target organ systemic toxicity—repeated exposure: ND
- ☐ Aspiration hazard: ND

3) 1-Butene, polymer with ethene

- ☐ Acute toxicity
 - o Oral: ND
 - o Skin: ND
 - o Inhalation: ND
- ☐ Skin corrosion/irritation: ND
- ☐ Serious eye damage/irritation: ND
- ☐ Respiratory sensitization: ND
- ☐ Skin sensitization: ND
- ☐ Carcinogenicity
 - o Occupation safety and health act: ND
 - o Notification of ministry of labor: ND
 - o IARC: ND
 - o OSHA: ND
 - o ACGIH: ND
 - o NTP: ND
 - o EU CLP: ND
- ☐ Germ cell mutagenicity: ND
- ☐ Reproductive toxicity: ND
- ☐ Specific target organ systemic toxicity—single exposure: ND
- ☐ Specific target organ systemic toxicity—repeated exposure: ND
- ☐ Aspiration hazard: ND

4) Talc, Non-asbestos form

- ☐ Acute toxicity
 - o Oral: ND
 - o Skin: ND
 - o Inhalation: ND
- ☐ Skin corrosion/irritation: 300µg/3days(Human), weak stimulus
- ☐ Serious eye damage/irritation: ND
- ☐ Respiratory sensitization: ND
- ☐ Skin sensitization: ND
- ☐ Carcinogenicity
 - o Occupation safety and health act: ND
 - o Notification of ministry of labor: ND
 - o IARC: ND
 - o OSHA: ND
 - o ACGIH: ND
 - o NTP: ND
 - o EU CLP: ND
- ☐ Germ cell mutagenicity: ND
- ☐ Reproductive toxicity: ND
- ☐ Specific target organ systemic toxicity–single exposure: ND
- ☐ Specific target organ systemic toxicity–repeated exposure: ND
- ☐ Aspiration hazard: ND

5) Second Antioxidant (2,4-DI-TERT-BUTYLPHENOL PHOSPHITE)

- ☐ Acute toxicity
 - o Oral: LD50 > 6000mg/kg Rat
 - o Skin: LD50 > 2000mg/kg Rat (GLP DATA)
 - o Inhalation: ND
- ☐ Skin corrosion/irritation: No Skin Irritation for Human
- ☐ Serious eye damage/irritation: Highly Irritate for Rabbit eyes
- ☐ Respiratory sensitization: ND
- ☐ Skin sensitization: No Skin Sensitization for Guinea Pig
- ☐ Carcinogenicity
 - o Occupation safety and health act: ND
 - o Notification of ministry of labor: ND
 - o IARC: ND
 - o OSHA: ND
 - o ACGIH: ND
 - o NTP: ND
 - o EU CLP: ND
- ☐ Germ cell mutagenicity: Negative for bacteria, hamster, mouse

- ☐ Reproductive toxicity: NOAEL 292.6mg/kg bw/day
- ☐ Specific target organ systemic toxicity–single exposure: ND
- ☐ Specific target organ systemic toxicity–repeated exposure: Rat, 13 weeks, NOAEL > 147mg/kg
- ☐ Aspiration hazard: ND

6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidiny Esters)

- ☐ Acute toxicity
 - ☐ Oral: LD50 > 2000mg/kg Rat
 - ☐ Skin: ND
 - ☐ Inhalation: LD50 > 5mg/m³ Rat, 7hr
- ☐ Skin corrosion/irritation: Rabbit, Normal irritation / Dermal LD50 (rat) > 5000mg/kg² / Irritative for guinea pig
- ☐ Serious eye damage/irritation: Highly Irritate for Rabbit
- ☐ Respiratory sensitization: ND
- ☐ Skin sensitization: ND
- ☐ Carcinogenicity
 - ☐ Occupation safety and health act: ND
 - ☐ Notification of ministry of labor: ND
 - ☐ IARC: ND
 - ☐ OSHA: ND
 - ☐ ACGIH: ND
 - ☐ NTP: ND
 - ☐ EU CLP: ND
- ☐ Germ cell mutagenicity: Negative (AMES TEST)
- ☐ Reproductive toxicity: ND
- ☐ Specific target organ systemic toxicity–single exposure: ND
- ☐ Specific target organ systemic toxicity–repeated exposure: Rat, Duodenum, NOAEL > 40mg/kg/day
- ☐ Aspiration hazard: ND

12. ECOLOGICAL INFORMATION

A. Aquatic, terrestrial organisms toxicity

- 1) Polypropylene: ND
- 2) Ethylene–Propylene copolymer: ND
- 3) 1–Butene, polymer with ethene: ND
- 4) Talc, Non–asbestos form

- ☐ Fish: LC50 > 10000 mg/ℓ 96 hr Brachydanio rerio
- ☐ Crustacea, algae: ND

5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE)

- ☐ Fish: LC50 > 42 mg/ℓ 96 hr
- ☐ Crustacea: LC50 > 510 mg/ℓ 24 hr
- ☐ Algae: LC50 > 95.2 mg/ℓ 72 hr

6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidiny Esters)

- ☐ Fish: LC50 > 42 mg/ℓ 96 hr

☐ Water flea: EC50 > 0.1 mg/ℓ 48 hr

☐ Algae: EC50 > 11 mg/ℓ 72 hr

☐ Bacteria: IC50 > 100 mg/ℓ 3 hr

B. Persistence and degradability

1) Polypropylene: ND

2) Ethylene–Propylene copolymer: ND

3) 1–Butene, polymer with ethene: ND

4) Talc, Non–asbestos form

☐ Persistence: log Kow –1.50 (Estimate)

☐ Degradability: ND

5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE)

☐ Persistence: > 6 (Calculation Value)

☐ Degradability: ND

6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidiny Esters): Not Biodegradable

C. Bioaccumulative potential

1) Polypropylene: ND

2) Ethylene–Propylene copolymer: ND

3) 1–Butene, polymer with ethene: ND

4) Talc, Non–asbestos form: ND

5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE)

☐ Condensability: BCF 4.66

☐ Biodegradability: 6%, 28 days

6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidiny Esters)

: BCF < 11 (in remarkable range) in carp

D. Mobility in soil: ND

E. Other adverse effects: ND

13. DISPOSAL CONSIDERATIONS

A. Disposal methods

☐ Landfill after incineration or incineration

B. Disposal considerations(Specify disposal container and methods)

☐ Incinerate over 800°C at incineration plant equipped dust collector

14. TRANSPORT INFORMATION

A. UN Number: Not applicable

B. UN Proper Shipping Name: Not applicable

C. Transport hazard class(es): Not applicable

D. Packing group, if applicable: Not applicable

E. Environmental hazards: Not applicable

F. Special precautions for user

☐ Cover with plastic film to prevent raining and fluttering

15. REGULATORY INFORMATION

A. Restriction according to occupation safety and health act

- 1) Polypropylene: ND
- 2) Ethylene–Propylene copolymer: ND
- 3) 1–Butene, polymer with ethene: ND
- 4) Talc, Non–asbestos form
 - ☐ Working environment measurement(cycle : 6 month)
 - ☐ TLVs(Threshold Limit Values) chemicals
- 5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE): ND
- 6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidinyl Esters): ND

B. Restriction according to toxic chemicals control act

- 1) Polypropylene: ND
- 2) Ethylene–Propylene copolymer: ND
- 3) 1–Butene, polymer with ethene: ND
- 4) Talc, Non–asbestos form: ND
- 5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE): ND
- 6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidinyl Esters): ND

C. Restriction according to safety control of dangerous substances act

- 1) Polypropylene: ND
- 2) Ethylene–Propylene copolymer: ND
- 3) 1–Butene, polymer with ethene: ND
- 4) Talc, Non–asbestos form: ND
- 5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE): ND
- 6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidinyl Esters): ND

D. Restriction according to wastes control act

- 1) Polypropylene: ND
- 2) Ethylene–Propylene copolymer: ND
- 3) 1–Butene, polymer with ethene: ND
- 4) Talc, Non–asbestos form: ND
- 5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE): ND
- 6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidinyl Esters): ND

E. Restriction according to other domestic and foreign law

- 1) Polypropylene: ND
- 2) Ethylene–Propylene copolymer: ND
- 3) 1–Butene, polymer with ethene: ND
- 4) Talc, Non–asbestos form: ND
- 5) Second Antioxidant (2,4–DI–TERT–BUTYLPHENOL PHOSPHITE): ND
- 6) Hindered Amine Compound (Fatty acids, 2,2,6,6–Tetra methyl–4–Piperidinyl Esters): ND
 - ☐ International Chemical Control Laws
 - ☐ Europe List: Pre–Registration

- o America List (TSCA): Registration
- o Japan List: Registration

16. OTHER INFORMATION

A. References and sources for data

- KOSHA CODE W-05-2008 [Guidelines for writing of MSDS, 2008. 6. 20]
- Practice for writing of KOSHA MSDS (Educational development institute of textbook 2008-9-70)
- Occupation safety and health act
- ICSC(International Chemical Safety Cards)
- HSDB(Hazardous Substances Data Bank)
- QSAR(Quantitative Structure Activity Relationships)
- NLM(National Library of Medicine)
- IUCLID(International Uniform Chemical Information Database)
- The Merck Index 13th Ed.
- Corporate Solution From Thomson Micromedex(<http://csi.micromedex.com>)

B. Originated date

Oct 10th 2023

C. Revision number and date

D. Etc.

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